Week 3 - Learning Map

Goal: Build a SupportDesk RAG assistant. Implement a baseline pipeline with retrieval and generation, add multi turn history, measure precision at K and groundedness, iterate using A versus B tests, then upgrade to agentic RAG with tool routing in LangChain safely.

Overview...

Week 3 takes you from basic retrieval building to a complete RAG assistant you can test and improve.

You will implement a full pipeline: offline indexing, online querying, top K retrieval, prompt context assembly, and answer generation.

You will add multi turn conversation support by tracking history and grounding each turn in retrieved evidence.

Then you will learn evaluation as a two layer framework. You will learn common RAG failure patterns.

For retrieval, you will measure Precision at K, Recall at K, and F1 at K, and understand how K trades precision for recall.

For generation, you will score groundedness, completeness, and answer relevance.

You will use an LLM as a judge with rubric prompts and temperature zero to produce repeatable scores, set thresholds, and run A versus B tests while watching latency and cost.

Finally, you will convert direct RAG into agentic RAG using the ReAct loop, strong tool descriptions, safe error handling, and multi step tool use in LangChain.

Note:

Post-class resources are optional and intended for post-class learning.

HOW TO START?

Roadmap

Pre-class Setup

- Clone repo + run Module 4/5/6 demos
- Recall Week 2: offline indexing foundation (embeddings, chunking, indexing)
- Target outcome: SupportDesk-RAG-Workshop

Live Class

Build the RAG Pipeline

- Online loop: query → retrieve Top-K → assemble context → generate answer
- Prompt/context formatting + citations
- Multi-turn RAG: conversation history
- LangChain components + LCEL composition

Evaluate + Iterate

- Two-layer eval: Retrieval + Generation
- Retrieval: Precision@K / Recall@K / F1@K and choosing K trade-offs
- Generation: groundedness, completeness, answer relevance
- LLM-as-Judge + thresholds + A/B pipeline

Learners are encouraged to follow along with the instructor during the live session and execute the code on their local environment.

Upgrade to Agentic RAG

- Direct RAG vs Agentic RAG (when multi-step)
- ReAct loop: Reason → Tool → Observe → loop
- Tool design + routing (3–7 tools)
- Best practices: errors, max iterations, cost/latency monitoring.

Post-class resources (Optional deep dives)

- Financial Bot for LLMs Project

Assignment

- Exercises - RAG Pipeline, Evaluation, Agentic RAG

Please use the Wednesday AMA session to clarify any doubts you have on the topics. The assignment solutions will be discussed during the Saturday Assignment Review session.

All the resources have been added to your Uplevel account. If you have any questions, please reach out to the Operations team or raise a support ticket.